

Original Report

Audit tool for external beam radiation therapy departments

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Abstract

Purpose: Development of a self-contained audit tool for external beam radiation therapy to assess compliance with the major recommendations from professional organizations and generally accepted standards of practice. Intensity modulated radiation therapy, stereotactic body radiation therapy, stereotactic radiosurgery, and volumetric modulated arc therapy were included in this review.

Methods and Materials: A physics quality working group developed a department vision, distinguished and summarized key references, and condensed important elements of good documentation practices. The results were then compiled in a checklist format and used to perform audits at 3 sites.

Results: The final audit tool contains 65 items spanning a wide range of external beam radiation therapy practices. Several of the audit items address issues not commonly identified by other authoritative sources. A total of 48 process improvements were identified at the 3 sites audited.

Conclusions: The enclosed self-inspection list may be useful to a site as an annual review tool, as an aid in preparation for the American College of Radiology-American Society for Therapeutic Radiology and Oncology practice accreditation, or as a catalyst for general quality improvement. Sites can quickly identify opportunities for improvement by concentrating on high importance items and commonly identified areas of noncompliance.

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Introduction

The field of radiation oncology is both technology and procedure intensive. From initial consult to patient treatment, there are literally hundreds of steps that involve the interplay of people, software, and sophisticated delivery systems. The very nature of the treatment, the delivery of potentially lethal doses of radiation if

administered improperly, warrants the use of carefully developed processes and quality assurance (QA) measures. The professional societies in radiation oncology have been making recommendations in this area for years, such as the report of the American Association of Physicists in Medicine (AAPM) Task Group 40¹ and special issues of the *International Journal of Radiation Oncology Biology and Physics* focused on quality.^{2,3} Recent news reports have brought increased focus on safety and quality with calls for additional scrutiny.⁴⁻⁸ As a whole, professionals in this field are embracing the challenge but may find themselves overwhelmed by the volume of information available to guide clinical practice. An example is the sub-specialty of external beam radiation therapy. There are more than a dozen organizations and

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governing bodies that have written tens of thousands of applicable pages of guidance documents, regulations, reports, and white papers. The American College of Radiology and American Society for Radiation Oncology (ACR–ASTRO) Practice Guidelines describe quality practices against which a facility is inspected if they pursue accreditation.⁹⁻¹⁵ The AAPM has published over 100 task group reports, many of which apply to external beam radiation therapy.¹⁶⁻²⁸ Although the Nuclear Regulatory Commission does not currently regulate X-ray producing machines and accelerator-produced isotopes, individual state regulations must be strictly followed. The Center for Radiation Control Program Directors (CRCPD) lists suggested state regulations for radiation oncology, which form the nucleus of the regulations in some states.²⁹ The International Atomic Energy Agency, International Commission on Radiation Units and Measurements, National Council on Radiation Protection and Measurements, Radiation Therapy Oncology Group, and other organizations have likewise produced excellent and detailed technical publications with quality and safety procedures applicable to the clinic. In addition, the peer-reviewed literature contains tens of thousands of additional pages with recommendations, and each equipment manufacturer has manuals, safety notices, customer technical bulletins, and a variety of documents describing quality practices. Sifting through these unabridged resources in detail is an impossible task.

Often a clinical physicist is assigned a task such as “ensuring quality,” a goal that can never be achieved by one person. In reality, a facility must create and nurture a culture of both quality and safety. Familiarity with and intelligent application of the above-mentioned references are 2 components of this culture.

In light of these challenges, a group of physicists set out to summarize key quality practices as part of an effort to spark an environment that places the highest emphasis on quality. The initial goal was to highlight the most critical changes in an easy-to-use format. The final product is intended as a peer review or self-assessment tool that may aid a site in identifying areas for improvement. It may be particularly useful to a small clinic or single physicist site. Many of the items reflect the authors’ interpretation of the listed references; therefore, in no way are the authors intending that this sample product be interpreted as a prescriptive document or even a finished work. Instead, it is hoped that sites will find the described approach helpful as they develop their own quality management program.

Methods and materials

A group of 5 board-certified radiation therapy physicists formed a team that met semiweekly over a period of

6 months. These 5 physicists work at 3 different facilities as part of an affiliated network of 8 radiation oncology clinics in the State of Michigan. The team members come from a varied background of training and experience and have worked within the same network for between 3 and 25 years. All sites use linear accelerators and information systems from a single vendor, as well as a common treatment planning platform. The group started with questions that we hoped to address: “What are minimum standards for quality that a site should achieve?” “If we were stepping into a radiation therapy clinic what quality processes and procedures would we like to see already in place?” “How do we make quality a primary clinical focus?” and “What constitutes good documentation?” Although significant time was spent on these preliminary discussions, they sowed the seeds that led to a written vision statement. This statement clearly outlined what the team believed the department should look like once the transformation to a culture of quality was matured. This full transformation will require active institutional endorsement, participation by the entire department, a foundation of documented practices and “due process,” and checks and balances that eliminate single points of failure, provide feedback, and ensure follow-up. The vision statement as finalized is listed below.

A department exemplifying an all-encompassing, patient-centered culture of quality includes the following.

- (1) Living, accessible, and approved documents that standardize the routine, clinically significant procedures.
- (2) Guidelines for identifying and managing exceptions to the standard of practice.
- (3) Concordance with the major recommendations from regulatory agencies, professional societies, and any state regulatory control agency.
- (4) Training and feedback for executing and continually improving department policies and procedures.
- (5) Regular and proactive validation of department quality measures through internal and external peer review.
- (6) Approved methods for acceptance testing and commissioning that are consistent with good clinical practice and professional society recommendations.
- (7) Staff dedicated to quality management to include adequate full-time support.
- (8) Continuous and active oversight of quality practices by a quality committee.

The team then turned their attention to identifying the major, concrete practices and processes that would be expected of facilities performing external beam radiation therapy to include intensity modulated radiation therapy, stereotactic body radiation therapy, stereotactic radiosurgery, and volumetric modulated arc therapy. Key guidance from the AAPM, ACR, ASTRO, and CRCPD were

identified and summarized. Important quality-related items were then compiled, and items with low severity and frequency were culled in order to make the first improvement initiative tractable. The compiled recommendations were then assembled in a checklist suitable as an audit or peer review tool. Emphasis was placed on meeting CRCPD guidelines (which often mirror the individual state requirements), incorporating the ACR-ASTRO recommendations that are considered essential for accreditation, and identifying vital physics and administrative responsibilities that enhance quality or guard against treatment variances. The format and content of documentation was also pinpointed as a crucial element in quality practice. As such, the key elements of a quality documentation program were identified. For this latter step the experiences of the authors, individual recommendations in the aforementioned guidelines, the major documentation practices as outlined by the International Organization for Standardization,³⁰ and the invited recommendations of an outside colleague (S. Meeks, PhD, written communication; September 7, 2010) were used to formulate basic recommendations.

The checklist was then tested at 3 different clinical sites that were part of the clinical network previously described. Each site was scrutinized for compliance with the checklist criteria. Results were used to identify potential process improvements, revise the audit list itself, and draw general conclusions on the types of noncompliances identified. The audits were performed by 3 different pairs of physicists, with all but one being members of the quality team that developed the checklist. Each item on the checklist was weighted for importance by a team of 3 designated physicists: “1” is a noncompliance with one of the team recommendations; “2” is a noncompliance with one of the authoritative recommendations; and “3” is a noncompliance with one of the authoritative recommendations that carries, subjectively, highly significant clinical implications. As a point of clarification all checklist items based on authoritative recommendations include the appropriate reference in the checklist. Each noncompliance was then scored by the same 3 designated physicists according to severity: “1” indicates the task was accomplished but not appropriately documented; “2” indicates the task was only partially accomplished or multiple deficiencies in execution and documentation were noted; while “3” indicates that the task was not accomplished or documented in any way. Use of such a scoring system indicating importance and severity allows an easy visual analysis of the audit’s impact.

To permit further generalization on the types of improvements identified, the noncompliances were also broken down according to whether they fell under the 3 broad categories of documentation or training, QA or organizational practice, or clinical practice. Each of these subtypes was further grouped under the categories of team recommendation or authoritative recommendation.

Results

Table e1 (available online only at www.practicalradonc.org) shows the detailed results of this self-improvement effort. Primary responsibility (PR, column 2) refers to the member of the department who may be best suited to assess the site’s compliance with the particular recommendation. He or she is not the party solely responsible for accomplishing that item. The next column, labeled “I,” shows the importance, on a scale of 1 to 3, given to that item using the criteria described earlier. The third column (standard-criteria) provides a brief summary of the item to be investigated. The balance between sufficient detail and the need for brevity posed a challenge when writing this summary. In some cases the team expanded and interpreted the guidance document to capture the spirit of the recommendation and provide concrete criteria. References (column 4) were therefore included whenever applicable, enabling the user to read and interpret the original guidance document in the event the criteria are unclear. They also allow the user to filter out the authors’ interpretation or application of the authoritative source. A number of the checklist items covered areas where no suitable reference was identified, and in those cases an attempt was made to ensure the item was self-explanatory. These items are referred to as team recommendations. For the working checklist, the authors also used “standard met” and “follow-up” columns, while readers may choose to simply track each item by number in a separate document or cut and paste the columns into a spreadsheet and customize as desired.

The results of the 3 test audits were summarized using bubble plots and a pie chart. The bubble plots demonstrate the importance of the item on the abscissa and the severity of the finding on the ordinate. The pie chart shows the breakdown of the audit findings using the previously described criteria. Checklist items that demonstrated noncompliances at 2 or more of the inspected sites include numbers 4, 9, 10, 11, 12, 13, 14, 16, 19, 32, 51, 52, and 55.

Discussion

The genesis of the working group resulted in a vision statement that parallels that found in other ASTRO, AAPM, and ACR documents. For example, 7 of the 8 items in the vision statement are addressed directly or indirectly by the proposed department environment found in Moran et al.³¹ Formulating an individualized vision statement provides the benefit of “buy in” to a culture of quality and ensures team members are focused on the same end goal.

The compiled self-inspection list may be useful to a clinic as a single-use catalyst for quality improvement,



Figure 1 Bubble plot showing the number of identified noncompliances at site 1 as a function of criteria importance and noncompliance severity.

a periodic or annual review tool, or as an aid in preparation for ACR-ASTRO accreditation. It traces current procedures and practices, extracts key items from professional organizations and published guidance documents, and aids in targeting weaknesses in documentation and record keeping. The abridged nature of this list, mandated by practical considerations, means that compliance with the itemized expectations represents only one step toward mature quality practices. As examples, no attempt was made to incorporate the requirements of specific states, and some special procedures (such as total skin electron therapy and total body irradiation) were omitted from consideration. It is hoped that individuals using this list will scrutinize and modify it as needed, especially if it is used repeatedly as part of annual review. Self-directed audits can and should be performed using a variety of sources and references, and the reader is referred to key documents for alternative approaches.^{1,16,32-34}

When running through the checklist it is imperative that the references be readily available because in many cases the audit items are simply pointers to crucial details in the reference.^{1,9-23,25-29,35-41} Access to the required documents is most easily accomplished through internet access. The amount of time needed to accomplish the audit will

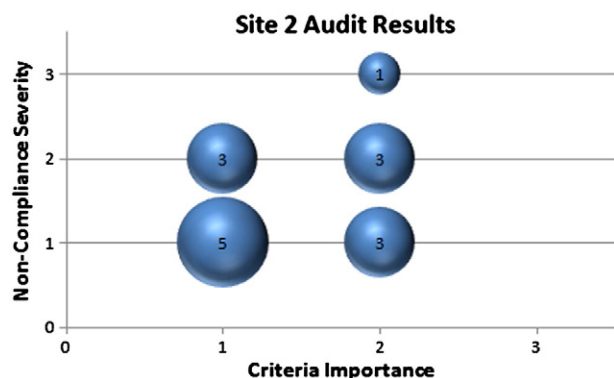


Figure 2 Bubble plot showing the number of identified noncompliances at site 2 as a function of criteria importance and noncompliance severity.

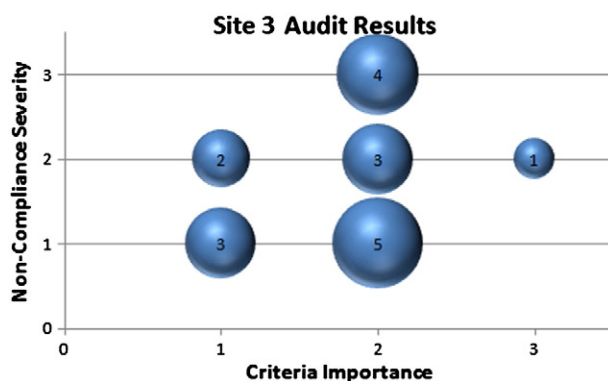


Figure 3 Bubble plot showing the number of identified noncompliances at site 3 as a function of criteria importance and noncompliance severity.

vary greatly and depend on the auditor's familiarity with both the audited site and the references. The 3 test audits were performed, with one exception, by the authors of the checklist, and each audit took 1 full day. It may be prudent for other users to allow more time than this, especially if preparatory activities such as reviewing references and scrutinizing site-specific policies and other documents are considered. Physician and administrative support are required to liberate the time to accomplish the review, assist with the audit itself, and promptly implement the resulting process improvements. The audit can serve to coordinate the team of administrator, physician, and medical physicist, not only to enable a smoother and more transparent treatment process but to foster awareness of each other's responsibilities and charges. Results are to be forwarded to the appropriate continuous quality improvement committee for consideration and action. To continue the culture of quality, appropriate and ongoing follow-up through the CQI committee itself, appropriate peer review, and department focus groups are needed.

Figures 1 through 3 show the resultant bubble plots for each audited site. The majority of the findings involved items with an importance of 1 and 2, while there was a wide range in the judged severity of the noncompliances. Such a scoring system is useful for prioritizing improvement and remediation efforts. Items with an importance of 3 were judged as an immediate priority, and those tasks were accomplished in reverse order of noncompliance severity. Findings with an importance of 2 are to be tackled next using the same severity ordering, then finally findings with an importance of 1.

A pie chart breakdown describing the areas where noncompliances were found is shown in Fig 4. From this chart it is evident that the bases for noncompliances were approximately evenly split between team and authoritative recommendations, providing validation that the team effort helped to identify new opportunities for process improvement among the affiliate sites. The types of findings were also evenly split between documentation-training and QA-organization practices. The small number



Figure 4 Pie chart showing a breakdown of the 48 non-compliances identified during the 3 site audits. Categories are described on the chart. QA, quality assurance.

of clinical practice findings was expected, reflecting the limited number of clinical practice items in the audit list. Because the development team consisted entirely of physicists, only clinical items mandated by an authoritative source were included. A similar list developed by a focus group of radiation oncologists and skewed toward clinical practices would be beneficial to the practice community.

For sites that want to quickly identify potential areas for improvement, an abbreviated audit is possible. Items with an importance of “3,” for example, could be inspected and acted on first without considering other categories. Another option would be to scrutinize the item numbers identified in the [Results section](#) as being noncompliant at multiple sites. This latter approach may be slightly skewed, as the test audits were performed at only 3 sites and these sites have ongoing collaborate efforts, but may nonetheless help to quickly identify weaknesses in practices or documentation.

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